

0 3

In Europe, viruses have infected a large number of frogs of different species. The viruses are closely related and all belong to the Ranavirus group.

Previously, the viruses infected only one species of frog.

0 3 . 1

Suggest and explain how the viruses became able to infect other species of frog.

[2 marks]

0 3 . 2

Name **two** techniques the scientists may have used when analysing viral DNA to determine that the viruses were closely related.

[1 mark]

1

2

0 3 . 3

Determining the genome of the viruses could allow scientists to develop a vaccine.

Explain how.

[2 marks]

03.4

Describe how the B lymphocytes of a frog would respond to vaccination against Ranavirus.

You can assume that the B lymphocytes of a frog respond in the same way as B lymphocytes of a human.

Do **not** include details of the cellular response in your answer.

[3 marks]

8

Turn over for the next question

